

365 Data Science SQL Basics – Topic 2.4: Aggregate Functions & Grouping

Exhaustive Notes on Aggregate Functions, GROUP BY, HAVING & WHERE vs HAVING

1. Aggregate Functions Overview

• `COUNT()`: Counts non-null records. • `SUM()`: Total sum of numeric column. • `AVG()`: Arithmetic mean. • `MIN()` / `MAX()`: Smallest and largest value.

2. GROUP BY & HAVING Clauses (WHERE vs HAVING)

• `GROUP BY`: Group rows sharing common attribute values. • `HAVING`: Filters aggregated groups after calculation (unlike `WHERE`, which filters raw rows before grouping).

```
SELECT department_id, COUNT(*) as total_emp, AVG(salary) as avg_sal
FROM employees
WHERE status = 'Active'
GROUP BY department_id
HAVING AVG(salary) > 60000;
```

3. Topic 2.4 Viva Q&A

1. Q: What is the main difference between WHERE and HAVING? A: `WHERE` filters rows before grouping; `HAVING` filters aggregated groups after `GROUP BY` execution.